

PHD RESEARCH ROADMAP

ACADEMIC YEAR 2024 / 2025

CompanionAI

Research Programme

Building Loneliness Detection and Embedded Intervention
Systems with Microsoft Azure

A

Data Lead

B

Detection Lead

C

Intervention Lead

D

Deployment Lead

RESOURCE PARTNER

STUDENTS A, B, C & D

COLLABORATIVE FACULTY RESEARCH

Powered by **Microsoft Azure**

Why Loneliness Matters

A global public health crisis meeting a critical technological vacuum.

 WHO DESIGNATION

Social isolation is a major risk factor equivalent to **smoking 15 cigarettes per day.**

"Loneliness is now recognized as a global public health priority."

 The Current Research Vacuum



Data Scarcity

No locally-grounded datasets capturing cultural, behavioral, and physiological signals.



Cloud Dependency

Most tools require constant internet/cloud access — not viable in low-resource settings.



Cultural Inflexibility

AI companions lack cultural adaptation for non-Western linguistic and social contexts.



Hardware Gap

Absence of affordable, real-time embedded systems capable of offline inference.

 The Accessibility Barrier

- Existing solutions are largely clinical and expensive.
- Inaccessible to underserved and low-resource populations.

 **CompanionAI addresses all four gaps through a single, integrated research ecosystem.**

What CompanionAI Will Build

A four-layer closed-loop system for accessible loneliness support.



LAYER 1: DATA

Multimodal Dataset

A rich, locally collected dataset capturing text, voice, physiological, and behavioral signals for culturally grounded ground truth.



LAYER 2: DETECTION

ML Intelligence

Real-time machine learning models that detect loneliness signals with high accuracy, optimized for limited data environments.



LAYER 3: INTERVENTION

AI Companion

A personalized, culturally adaptive conversational AI delivering evidence-based responses grounded in therapeutic principles.



LAYER 4: DEPLOYMENT

Embedded System

Lightweight Arduino/mobile platforms running all layers affordably, offline-capable, and at scale for real-world impact.



PROGRAMME END GOAL

An open, reproducible, and ethical AI framework that reduces loneliness through accessible embedded technology.

POWERED
BY

Azure

The Central Research Question

DEFINING THE SCOPE OF THE PROGRAMME

“ How can a locally constructed multimodal loneliness dataset, combined with culturally adaptive machine learning models, be used to build and deploy an affordable, real-time embedded AI companion system that measurably reduces loneliness in target populations?

STUDENT A: DATA



How do we build a valid, locally grounded multimodal loneliness dataset?

FOUNDATION

STUDENT B: DETECTION



How do we detect loneliness accurately from multimodal signals using lightweight ML?

INTELLIGENCE

STUDENT C: INTERVENTION



How do we design and evaluate personalized AI conversations that reduce loneliness?

INTERACTION

STUDENT D: DEPLOYMENT

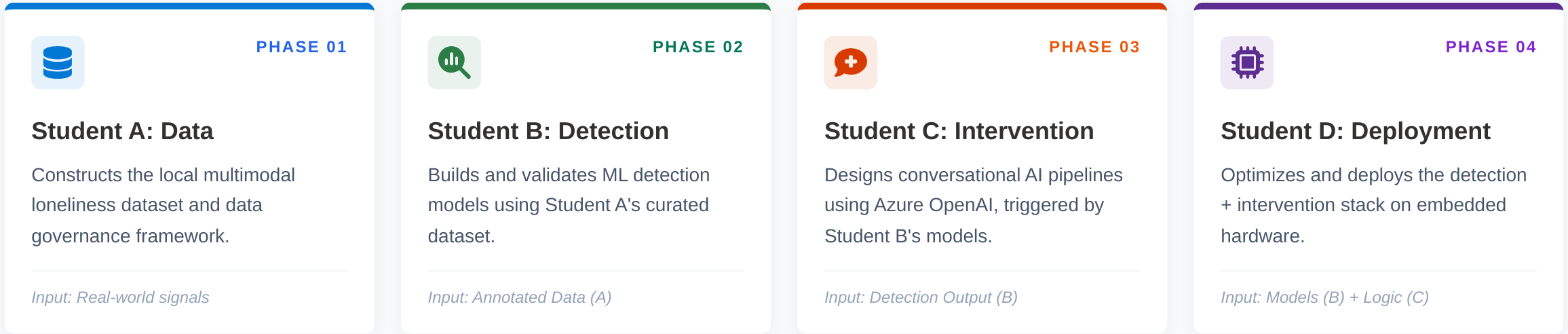


How do we deploy the full CompanionAI system on an embedded device under real-world constraints?

IMPLEMENTATION

The Programme Logic

A Closed-Loop Multi-Student Research Ecosystem



Continuous System Enrichment (The Loop)

Student D's deployed system feeds real-world interaction data back to Student A for re-annotation and model refinement.

DATA → DETECTION → INTERVENTION → DEPLOYMENT

Our Microsoft Azure Stack

The foundational infrastructure for the CompanionAI ecosystem.

DATA & GOVERNANCE LAYER

Azure Data Lake Storage Gen2 Azure Blob Storage Azure Purview (Governance)

Azure Data Factory (Pipelines)

Centralized, secure storage with full lineage tracking.

AI & MACHINE LEARNING LAYER

Azure Machine Learning Studio Azure AutoML Cognitive Services (Speech/Emotion)

Azure OpenAI Service (GPT-4o) Language Understanding (LUIS)

Scalable model training, fine-tuning, and inference.

EDGE & IOT LAYER

Azure IoT Hub Azure IoT Edge Azure RTOS (Arduino) Azure Sphere (Security)

Secure device management and TinyML deployment.

DEVOPS & COLLABORATION LAYER

Azure DevOps GitHub Actions Azure Container Registry Azure Monitor

Unified CI/CD pipelines and experiment tracking.

 **Shared Resource Governance:** Single subscription with RBAC and a shared experiment registry for all four PhD students.

Data Lead

PHD RESEARCH TITLE

"Constructing a Locally Grounded Multimodal Dataset for Loneliness Detection in Low-Resource and Culturally Diverse Populations"

PHD RESEARCH QUESTION

"How can a valid, ethically governed, and culturally representative multimodal dataset capturing linguistic, vocal, physiological, and behavioral markers of loneliness be constructed for AI model training in locally specific contexts?"

Role in CompanionAI

Student A is the **data foundation** of the entire programme. Without this high-quality local dataset, detection models, conversational evaluation, and hardware deployment lack a reliable ground truth.

Why this is PhD-level

Beyond simple data collection, the novelty lies in the **design and validation** of a specialized framework:

- ✓ Context-Aware Design
- ✓ Privacy-Preserving Architecture
- ✓ Ethically Governed Annotation
- ✓ Non-Western Population Focus

Student A | Literature to Read

Progressive Reading Roadmap: Data Engineering & Ethics

DATA LEAD



Foundational

CONCEPTS & ETHICS

- **Psychology of Loneliness:** UCLA & De Jong Gierveld Scales.
- **Multimodal Intro:** Text, speech, physiology, and behavioral logs.
- **AI Ethics:** IRB protocols, consent, and privacy-by-design.



Intermediate

METHODS & TOOLS

- **Annotation Methods:** Inter-annotator agreement & labeling.
- **Affective Corpora:** IEMOCAP, MSP-Podcast & local gaps.
- **NLP for Health:** Depression & loneliness detection (ACL/EMNLP).
- **Azure Purview:** Governance, catalogs, and lineage.



Advanced

FRONTIER RESEARCH

- **Multimodal Fusion:** Annotation challenges for affective computing.
- **Privacy-Preserving:** Differential privacy & Azure Confidential Computing.
- **Dataset Documentation:** Datasheets for Datasets (Gebru et al.).
- **Longitudinal Design:** Tracking behavioral signals over time.

Dataset Creation Responsibilities

1



Design

MONTH 1

- ✓ Define loneliness construct via UCLA/De Jong Gierveld scales.
- ✓ Multimodal protocol: Text, Voice, Physiological, Behavioral.
- ✓ IRB submission and ethics protocols.
- ✓ Recruit culturally representative cohorts.

2



Collection & Annotation

MONTHS 2-3

- ✓ Deploy **Azure Data Factory** pipelines for ingestion.
- ✓ **Azure ML Labeling** for multimodal annotation.
- ✓ Compute inter-annotator agreement (IAA).
- ✓ Anonymization & Differential Privacy application.

3



Curation & Release

MONTHS 4-6

- ✓ Quality checks: Deduplication and outlier removal.
- ✓ Build data cards/lineage in **Azure Purview**.
- ✓ Deliver benchmark subsets to Students B and C.
- ✓ Publish dataset with full documentation.



KEY AZURE TOOLS FOR STUDENT A

Azure Data Lake Storage Gen2

Azure Data Factory

Azure ML Studio Labeling

Azure Purview

Azure Confidential Computing

Research Novelty & Outputs

Scientific Novelty

*"The first ethically governed, locally collected, multimodal loneliness dataset with culturally grounded annotation rubrics—designed specifically for AI model training in **non-Western, low-resource contexts**."*

KEY DELIVERABLES

CompanionAI Dataset

Multimodal, annotated, and ethically governed corpus.

Data Cards

Standardized documentation for transparency.

Governance Framework

Privacy-preserving ethics and IRB protocols.

Azure Architecture

Scalable data engineering pipeline blueprints.

TARGET VENUES

ACL

EMNLP

NeurIPS (Datasets)

LREC-COLING

FAccT

Mental Health NLP

Focus: Fairness, Accountability, and Transparency in Affective Computing.

Student B | Detection Lead

"Lightweight Multimodal Machine Learning Models for Real-Time Loneliness Signal Detection on Resource-Constrained Devices"

 RESEARCH QUESTION

How can multimodal ML models (text, speech, physiological) be designed, trained, and compressed to accurately detect loneliness in real time on resource-constrained edge devices while maintaining performance?

 ROLE IN COMPANIONAI

Student B converts Student A's dataset into detection intelligence. These models must be accurate enough for clinical relevance yet small enough to run on Arduino/mobile hardware for Student D's deployment.

 SCIENTIFIC NOVELTY

The systematic study of model compression (quantization/pruning) and multimodal fusion trade-offs specifically for loneliness detection, where misclassification has real-world human consequences.

Student B | Literature Review

DETECTION LEAD & ML ARCHITECTURE



SCHOLARLY ROADMAP

LEVEL 01: FOUNDATIONAL



Classical ML for Affect

Baselines: SVM, Random Forest, Logistic Regression for sentiment.



Deep Learning for NLP

Transformers: BERT, RoBERTa, DistilBERT for affective state detection.



Speech Feature Extraction

Signal Processing: MFCC, mel spectrograms, and openSMILE feature sets.

LEVEL 02: INTERMEDIATE



Multimodal Fusion

Early/Late fusion and cross-modal attention architectures.



Model Compression

Pruning, quantization, and knowledge distillation fundamentals.



TinyML Fundamentals

Deploying ML on microcontrollers (Warden's TinyML framework).



Azure ML Services

Experiment tracking, AutoML, and model registry documentation.

LEVEL 03: ADVANCED



Federated Learning

Training on sensitive health signals without centralizing data.



Uncertainty Quantification

Reliability metrics for mental health ML decision making.



Edge ML Deployment

Azure IoT Edge and ONNX model export for embedded targets.

Model Development Responsibilities

01 Baseline Modeling

MONTH 2

- 🧠 Establish classical ML baselines (SVM, Random Forest) on Student A's data.
- 📄 Extract text (BERT), speech (MFCC), and physiological (HRV) features.
- 📈 Train and evaluate unimodal models to identify performance gaps.

02 Multimodal Fusion

MONTH 3

- 🔗 Compare Early Fusion, Late Fusion, and Cross-modal Attention architectures.
- 🧪 Utilize **Azure ML Studio** for experiment tracking and hyperparameter tuning.
- ✅ Select the optimal fusion strategy for real-time loneliness detection.

03 Compression & Edge

MONTHS 4 - 5

- ⚙️ Apply **INT8/FP16 quantization**, pruning, and knowledge distillation.
- 📦 Export to **ONNX format** for hardware-agnostic deployment on TinyML.
- 🔗 Deliver compressed artefacts to Student D for embedded integration.

Research Novelty & Outputs

Research Novelty

"A systematic multimodal fusion and compression science for loneliness detection — establishing which combination of modalities and compression techniques preserves clinically meaningful detection accuracy at edge-device scale."

- ✓ Reusable model recipes beyond single classifiers.
- ✓ Quantification of clinical accuracy vs. hardware constraints.

Key Deliverables



Compressed ONNX Artefacts



Baseline & Fusion Model Suite



Fusion Comparison Report



Compression Trade-off Analysis



Experiment Registry: Azure Machine Learning Studio

Target Publication Venues

INTERSPEECH

ICASSP

ACL (Mental Health NLP)

IEEE Transactions on Affective Computing

EMNLP

NeurIPS ML4H Workshop

Student C | Intervention Lead

SECTION 4 — INTERVENTION RESEARCH



PHD RESEARCH TITLE

"Culturally Adaptive Personalized Conversational AI Interventions for Loneliness Reduction: Design, Evaluation, and Ethical Governance"

? PhD Research Question

"How can a personalized, culturally adaptive conversational AI companion — built on large language models — be designed, fine-tuned, and rigorously evaluated to deliver evidence-based interventions that measurably reduce loneliness in target populations?"



ROLE IN COMPANIONAI

The voice and personality of the system. Student C transforms detection signals from Student B into high-quality, culturally appropriate, and therapeutically effective conversational responses.



WHY THIS IS PHD-LEVEL

Novelty lies in the construction of an evaluation framework for AI-driven conversation impact on loneliness, specifically testing how cultural adaptation influences therapeutic outcomes in HCI.

Literature Roadmap: Intervention Science

A curated progression from foundational psychology to advanced AI safety and evaluation.

ROADMAP FLOW

 SIMPLE

 INTERMEDIATE

 COMPLEX

Foundational

- 01 CBT & Behavioral Activation:** Core evidence base for loneliness interventions.
- 02 Conversational AI:** Dialogue systems, intent detection, and response generation.
- 03 LLM Basics:** GPT architecture, prompt engineering, and fine-tuning fundamentals.

Intermediate

- 04 Empathic Dialogue:** Emotionally intelligent chatbots and empathy modeling.
- 05 Cultural Competence:** Adapting AI responses to diverse cultural expressions of loneliness.
- 06 Azure AI Stack:** Azure OpenAI Service, RAG with Azure AI Search, and BLEU/ROUGE metrics.

Advanced

- 07 Safety & Ethics:** Fine-tuning for harm avoidance and Azure AI Content Safety.
- 08 Human-in-the-loop:** Designing clinical validity and ethical oversight mechanisms.
- 09 Longitudinal Studies:** Measuring impact over time using validated loneliness scales.

Student C | System Responsibilities



MONTH 1

Framework Design

- ✓ Define intervention framework based on **CBT & Behavioral Activation**.
- ✓ Map loneliness signals to conversational response strategies.
- ✓ Establish cultural adaptation parameters (register, metaphors).



MONTHS 2–3

System Development

- ✓ Configure **Azure OpenAI (GPT-4o)** with therapeutic system prompts.
- ✓ Build **RAG Pipeline** using Azure AI Search for grounded knowledge.
- ✓ Implement **Content Safety** filters and detection triggers.



MONTHS 4–6

Evaluation & Refinement

- ✓ Conduct **Human-in-the-loop** evaluation with clinical psychology input.
- ✓ Run longitudinal studies using **UCLA Loneliness Scales**.
- ✓ Compare culturally adapted vs. generic responses for outcome delta.

Research Novelty and Outputs

💡 Scientific Novelty

"The first culturally adaptive conversational AI intervention framework specifically designed and clinically evaluated for loneliness reduction."

THE THREE PILLARS OF INNOVATION



RAG Grounding

Therapeutic dialogue anchored in CBT and Social Prescribing knowledge bases.



Cultural Science

Adaptation of language register, metaphors, and social context for target groups.



Rigorous HCI

Longitudinal outcome measurement using UCLA & De Jong Gierveld scales.

📁 Key Deliverables

✓ Intervention Pipeline

End-to-end Azure OpenAI & AI Search implementation.

✓ Cultural Framework

Documentation for adapting LLM responses to local contexts.

✓ Evaluation Protocol

Rubrics for conversation quality and clinical outcome measurement.

🚀 Publication Venues

ACL

EMNLP

CHI

CSCW

JMIR

NeurIPS ML4H

FAccT

Student D | Research Introduction

PHD RESEARCH TITLE

"Quality-Aware TinyML Deployment of a Multimodal Loneliness Companion System on Resource-Constrained Embedded Platforms (Arduino and Mobile)"

? CORE RESEARCH QUESTION

"How can a full-stack AI loneliness companion system be optimized and deployed on resource-constrained embedded hardware while preserving acceptable detection accuracy and intervention quality under real-world operational constraints?"

🏠 Role in CompanionAI

Student D acts as the **bridge between research and reality**, porting detection models and intervention pipelines to physical devices.



Arduino



Mobile

Offline-First

💡 Why this is PhD-Level

The novelty lies in the **quality-aware systems study**: analyzing how deployment optimizations (quantization, pruning, caching) impact the overall **therapeutic effectiveness** of the system.

Literature to Read

From Embedded Fundamentals to Quality-Aware Deployment



1. Foundational

- 01 Embedded Systems:** Microcontrollers, sensors, Arduino, power constraints.
- 02 Mobile AI:** Android/iOS ML frameworks (TFLite, Core ML).
- 03 TinyML Intro:** Pete Warden's fundamentals on deploying NNs on MCUs.



2. Intermediate

- 04 Model Optimization:** INT8/INT4 quantization, pruning, NAS for edge.
- 05 Azure IoT Stack:** Hub, Edge, RTOS, and Sphere security.
- 06 Offline-Capable AI:** Local inference, caching, and graceful degradation.



3. Advanced

- 07 HCI for Companions:** Form factor, haptics, and voice interfaces.
- 08 Edge Security:** Encrypted inference and Azure Defender for IoT.
- 09 Quality-Accuracy Science:** Compression effects on therapeutic outcomes.
- 10 Power Profiling:** Battery-life optimization for always-on AI.

Student D | Embedded Deployment Responsibilities

01

Environment Setup

MONTH 1

📁

Register devices via **Azure IoT Hub** & telemetry pipelines.

🔧

Profile constraints: SRAM, Flash, Clock speed, & Power budget.

🔗

Define On-device vs. Edge Cloud vs. Full Cloud architecture.

02

Model Porting & Opt.

MONTHS 2–4

📄

Convert ONNX to **TFLite/TinyML** with hardware-aware quantization.

🔊

Implement real-time sensor data acquisition (Voice/Behavioral).

🔗

Embed Student C's pipeline with **offline intent fallbacks**.

03

Integration & Field Testing

MONTHS 4–6

📺

Run field tests: Latency profiling, battery-life, and accuracy.

📝

Document **Quality-Accuracy trade-offs** from compression.

🔗

Deliver open-source TinyML artefacts & hardware blueprints.

☁️

KEY AZURE TOOLS FOR DEPLOYMENT

Azure IoT Hub

Azure IoT Edge

Azure RTOS

Azure Sphere

Azure Defender for IoT

Azure Monitor

TensorFlow Lite

Student D | Research Novelty and Outputs

Embedded Systems & Deployment Lead

SECTION 5

💡 Core Research Novelty

*"The first **quality-aware embedded deployment study** for a full-stack AI loneliness companion."*

CHARACTERIZING KEY TRADE-OFFS



TinyML Compression
Pruning & Quantization impact

Offline Capability
Edge-Cloud synchronization



Power Efficiency
Always-on vs. Battery life



Therapeutic Quality
Outcome-accuracy preservation

📁 Key Deliverables

- CompanionAI Embedded Prototype**
Full-stack Arduino + Mobile integrated system
- Deployment Blueprints**
Hardware schematics & TinyML serving architecture
- Benchmark Reports**
Latency, power, and outcome quality analysis
- Open-Source Artefacts**
Optimized TinyML models & Azure IoT documentation

🚀 Publication Venues

- IEEE TBME
- ACM CHI
- IEEE ESL
- MLSYS
- NEURIPS ML4H
- EMNLP SYSTEMS

What All Four Students Share

The Collective Foundation: Shared Research Infrastructure & Governance



Unified Dataset

CompanionAI Loneliness Dataset: Owned by Student A, utilized as the ground truth for models, intervention context, and deployment testing across the programme.



Azure Cloud Ecosystem

Single subscription with Role-Based Access Control (RBAC). A shared experiment registry in Azure ML Studio tracks all training runs and model versions.



Evaluation Protocol

Common outcome measures: UCLA and De Jong Gierveld scales are standard across all experiments to ensure cross-comparability of results.



Ethics Framework

Unified IRB approval and consent protocols. Data governance policies designed by Student A are binding for all team members.



Scientific Standards

Mandatory usage of Data Cards and Model Cards for transparency. Azure Purview tracks data lineage from collection to deployment.



Integration Syncs

Monthly integration meetings to resolve dependencies between data collection, model training, intervention design, and hardware deployment.

Six-Month Execution Timeline

Integrated Parallel PhD Research Tracks & Milestones

 **PHASE 1: FOUNDATION TO DELIVERY**

	M1: FOUNDATION	M2: DATA/BASELINES	M3: CORE DEV	M4: INTEGRATION	M5: WRITING	M6: DELIVERY
A	Ethics, recruitment, protocol design.	Data collection starts; Azure Labeling workflow.	M1 cohort annotation; deliver subsets to B/C.	Dataset refinement; final data cards.	Paper drafting & code documentation.	Full Dataset & Governance Release.
B	Literature review & Azure ML setup.	Unimodal baseline models; AutoML runs.	Multimodal fusion architecture selection.	Compressed models delivered to Student D.	Reproducibility packages & model cards.	Final Model Suite & Benchmark Report.
C	Intervention design & Azure OpenAI setup.	Knowledge base & RAG pipeline config.	Prototype conversational pipeline; adaptation.	Human-in-the-loop evaluation; expert review.	Longitudinal outcome evaluation (UCLA scale).	Final Evaluation & Effectiveness Analysis.
D	Azure IoT Hub & hardware profiling.	TinyML env ready; sensor data acquisition.	Hardware-aware compression & porting.	Full CompanionAI stack assembled on device.	Field tests: Latency & power profiling.	Open-Source Prototype & Blueprints.

 **DATA MILESTONE (M3)**

Annotated benchmark data transferred from A to B and C for modeling.

 **MODEL MILESTONE (M4)**

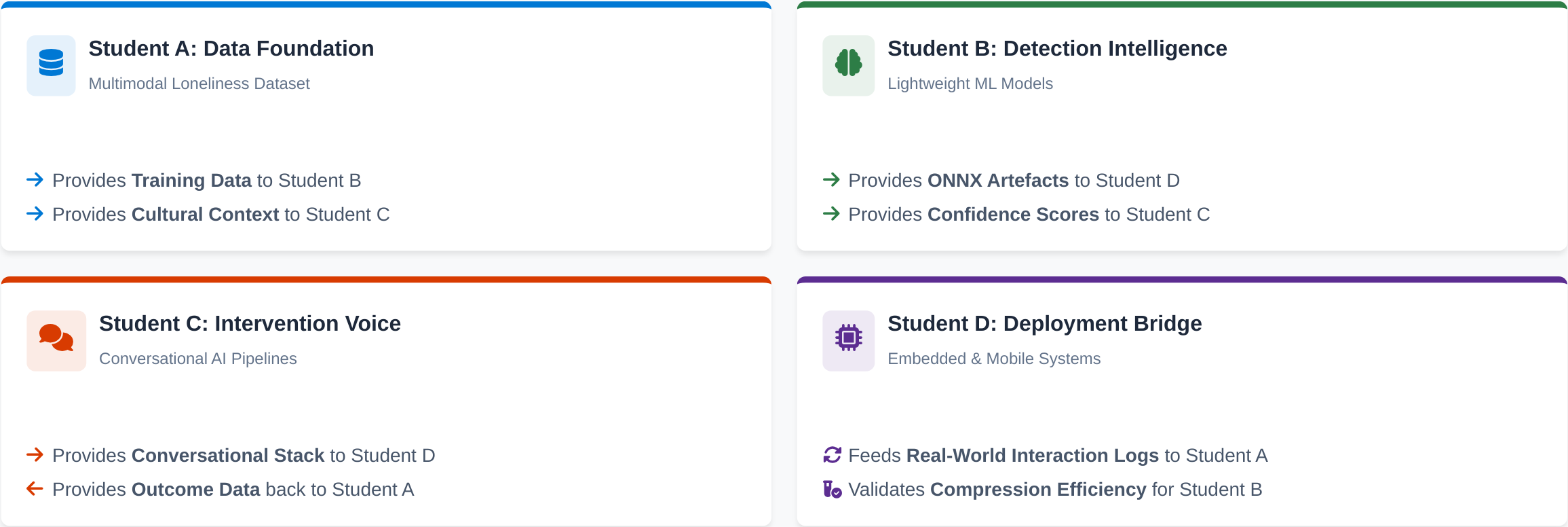
Compressed ONNX models delivered from B to D for embedded integration.

 **SYSTEM RELEASE (M6)**

Consolidated release of dataset, models, and hardware blueprints on GitHub.

How the Four Research Streams Connect

The Integrated Closed-Loop Architecture of CompanionAI





One Open CompanionAI Framework

Open-source, reproducible, deployable, and ethically governed.

RESEARCH LOOP

MULTI-MODAL

EDGE-NATIVE



Responsible AI at the Core of CompanionAI

Handling sensitive mental health data requires an ethical-first research framework. **Responsible AI is a research requirement, not an option.**

**Data Ethics**

- Mandatory IRB/Ethics approval
- Strict informed consent protocols
- Continuous right to withdraw

**Privacy**

- Azure Confidential Computing
- Differential privacy in ML training
- Data minimization principles

**Fairness**

- Demographic fairness evaluation
- Cultural adaptation testing
- Bias mitigation in model weights

**Safety**

- Azure AI Content Safety filtering
- Clinical escalation protocols
- Explicit non-replacement for care

**Transparency**

- Data Cards and Model Cards
- Open-source code and artifacts
- Reproducibility documentation

**Governance**

- Azure Purview for data lineage
- Bi-monthly ethics review cycles
- Role-based Access Control (RBAC)



What Success Looks Like

Measuring the integrated impact of the CompanionAI Research Programme.

STUDENT A ✓

The Data Foundation

Validation of the first ethically governed, locally collected multimodal loneliness dataset with culturally grounded rubrics and privacy-preserving architecture.

STUDENT B ✓

The Intelligence Core

Deployment of compressed, edge-ready detection models with a systematic science of performance–compression trade-offs for loneliness markers.

STUDENT C ✓

The Intervention Voice

A culturally adaptive conversational AI companion delivering evidence-based interventions with measured longitudinal reduction in loneliness scales.

STUDENT D ✓

The Real-World Bridge

A working embedded CompanionAI prototype (Arduino/Mobile) with open-source deployment blueprints and power/latency optimizations.



Programme-Level Milestones

Collective contributions to the global research community



High-Impact Publications (ACL, NeurIPS, CHI)



Open-Source CompanionAI Framework



Reproducible Azure Architecture

CompanionAI: No One Should Face Loneliness Alone

— and neither should this research.

FOUR PHD RESEARCH STREAMS. ONE INTEGRATED SYSTEM. ONE SHARED MISSION.



Student A

Building the **foundation** of understanding through grounded data.



Student B

Building the **intelligence** of detection with lightweight ML.



Student C

Building the **warmth** of conversation through adaptive AI.



Student D

Building the **reach** of deployment with embedded TinyML.



OPEN SCIENCE



BUILT FOR PEOPLE



ETHICALLY GOVERNED

RESEARCH PROGRAMME POWERED BY

Microsoft **Azure**